

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 2
 - Suprema and Infima

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Completeness Axiom

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- 1 is the supremum of

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- π is the supremum of

$$\{3, 3.1, 3.14, 3.141, 3.1415, 3.14159, 3.141592, \dots\}.$$

Challenge

Problem

Prove that if A is a set of integers that is bounded above, then A has a maximum.

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Solution

The set A is bounded above, so it has a supremum $b = \sup(A)$.

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Therefore $k \leq 0$ and $a' \leq a$.

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Therefore $k \leq 0$ and $a' \leq a$.

It follows that a is an upper bound of A , and since $a \in A$ it is a maximum.

Lower bounds and infima

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In other words

an infimum is a **greatest lower bound**

Properties of Suprema

The first property of suprema is that they must be arbitrarily close to elements of the set.

Theorem (Approximation Property)

Let $S \subseteq \mathbb{R}$ be bounded above, and let $b = \sup(S)$. Then for all $\epsilon > 0$, there exists $a \in S$ with

$$b - \epsilon < a \leq b.$$

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Proof.

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Since $b - \epsilon < b$, the definition of a supremum implies $b - \epsilon$ cannot be an upper bound.

Therefore there must exist $a \in S$ with $a > b - \epsilon$. □

Important example: The number e

For each n , let s_n denote the sum

$$s_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}.$$

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For every n ,

$$s_n \leq 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^n} = 3 - \frac{1}{2^n} < 3$$

so S is bounded above by 3.

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so S is bounded above by 3. Therefore S has a supremum, $\sup(S)$.

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We've just shown that $e^1 = e$ exists.

Properties of Suprema

Also, suprema play nicely with addition.

Theorem (Additive Property)

Let $A, B \subseteq \mathbb{R}$ be bounded above set

$$C = \{x + y : x \in A, y \in B\}.$$

Then C is bounded above and

$$\sup(C) = \sup(A) + \sup(B).$$

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Therefore $a + b \leq c$. □

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Then $b - 1 < b$ by Axiom 7, so $b - 1$ cannot be an upper bound.

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It follows from Axiom 7 that $b < n + 1$.

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This means there exists $n \in \mathbb{Z}_+$ with $b - 1 < n$.

It follows from Axiom 7 that $b < n + 1$.

However, $n + 1 \in \mathbb{Z}$, so this contradicts b being an upper bound.

Archimedian property

Theorem (Apostol Theorem 1.18)

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For every $x \in \mathbb{R}$, there exists $n \in \mathbb{Z}_+$ with $n > x$.

Proof.

If not, then x is an upper bound of $\mathbb{Z}_+$. □

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Theorem (Archimedean Property of Reals)

For every $x, y \in \mathbb{R}$ with $x > 0$, there exists $n \in \mathbb{Z}_+$ with $y < nx$.

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Theorem (Archimedean Property of Reals)

For every $x, y \in \mathbb{R}$ with $x > 0$, there exists $n \in \mathbb{Z}_+$ with $y < nx$.

Proof.

Replace x with y/x in the previous theorem. $\square$